# FLOOD INTELLIGENCE 3D

**Technical Design & Architecture Document | Version 2.0.0-PROD**

Live Deployment: https://flood-intelligence-3d-cy4j.vercel.app | Open-Source MIT License

## 1. Executive Summary & Problem Formulation

Urban and river basin inundation presents a critical spatiotemporal challenge where traditional optical satellites fail due to dense cloud cover, heavy rainfall, and nighttime limitations.

This platform couples active Synthetic Aperture Radar (SAR) with hydrodynamic physical constraints to achieve cloud-penetrating flood segmentation, continuous flood depth modeling (h), and real-time interactive 3D WebGL bathymetry.

## 2. System Architecture & Ingestion Pipeline

• Earth Observation Layer: Sentinel-1 C-SAR IW GRDH (VV + VH Dual-Pol radar backscatter), Copernicus DEM GLO-30 (30m elevation), and ECMWF ERA5 precipitation reanalysis.

• Radar Calibration: Radiometric sigma0 calibration and Refined Lee 7x7 speckle reduction to eliminate noise and urban corner reflections.

• Dual-Stage Machine Learning: U-Net++ with ResNet-34 backbone for flood extent segmentation coupled with a Physics-Informed Neural Network (PINN).

• Hydrodynamic Formulation: Solves 2D Shallow Water Navier-Stokes and Manning's roughness equations: dh/dt + div(uh) = R(t), with physical residual loss <= 4.18e-5.

## 3. 3D WebGL Visualization & Bathymetry Engine

• NASA Blue Marble Global View: Interactive 3D globe rendered with Three.js and React Three Fiber (R3F) featuring dynamic event markers.

• 3D Bathymetry Viewer: Terrain mesh procedurally displaced by Copernicus DEM elevation vertices with real-time wave simulation and layer toggles.

• Dynamic Cross-Section Profiler: Interactive 2D elevation cut-through tool displaying precise water depth above terrain bedrock for safe emergency navigation.

## 4. Enterprise Security & Authentication

• Real-Time 2FA OTP: New account registrations dispatch dynamic 6-digit codes via official Google SMTP (smtp.gmail.com:465) with 10-minute expiry.

• Direct Password Login: Returning sessions verify SHA-256 salted credentials directly without repeating OTP prompts.

• Brute-Force Countermeasures: Security locks triggered after 5 failed attempts with automated attempt resetting.

## 5. Automated Reporting & Spec-Compliant Export

• 10-Page Emergency Assessment Dossiers: Generates military/UN-grade PDF reports with hydrograph curves, risk matrices, and evacuation routes via jsPDF.

• Spec-Compliant TIFF 6.0 RGB Export: Downloads full-resolution GeoTIFF rasters natively viewable in Windows Photo Viewer and GIS suites (QGIS/ArcGIS).

## 6. Verification & Automated Test Coverage

• Automated test suite (npm test) verifies SHA-256 hashing determinism, 6-digit cryptographic entropy, expiration enforcement, and brute-force lockout with 100% pass rate (4/4 tests passed).